

Dynamic Regular Witnesses for Near-Exact State Complexity of Explicit Binary Consensus with Periodic Time

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September 6, 2026
Version v1.0.0 – not peer reviewed

Abstract

We study deterministic binary consensus with explicit termination in anonymous synchronous 1-interval-connected dynamic networks under one-bit broadcast-counting communication. Each round t exposes a free globally aligned phase $\phi_t = t \bmod P$, while the full round number is unavailable. Let $S_n(n, P)$ denote the minimum number of persistent states required when the network size n and the phase period P are known non-uniformly.

A direct predecessor obtained the known-size bounds

$$2 + \left\lceil \frac{\lfloor n/2 \rfloor}{P} \right\rceil \leq S_n(n, P) \leq 2 + \left\lceil \frac{n-1}{P} \right\rceil,$$

leaving a factor-two gap in the numerator. We reduce this to an additive constant. The main idea is a *dynamic regular witness*: every communication snapshot is regular, so opposite homogeneous executions follow topology-independent phase-augmented state orbits, while the topology is rearranged to make two backward causal cones grow almost one vertex per round. Using explicit quartic gadgets, for every $n \geq 15$,

$$2 + \left\lceil \frac{n-13}{P} \right\rceil \leq S_n(n, P) \leq 2 + \left\lceil \frac{n-1}{P} \right\rceil.$$

For even $n \geq 12$, cubic witnesses improve the lower numerator from $n-13$ to $n-10$. Thus, in the unsaturated regime where $n/P \rightarrow \infty$, the leading coefficient of the known-size state complexity is one.

We also determine the exact saturation threshold for the minimum possible three persistent states:

$$S_n(n, P) = 3 \iff P \geq n-1 \quad (n \geq 4).$$

The lower direction uses only a static path and combines anonymous path symmetry, a forced maximal-speed zero wave, and the binary broadcast alphabet. The claims are restricted to the stated model. We do not claim the full exact formula $S_n(n, P) = 2 + \lceil (n-1)/P \rceil$ for arbitrary state budgets.

Keywords. anonymous dynamic networks; binary consensus; explicit termination; state complexity; periodic time; one-bit communication; regular dynamic graphs; causal cones; local memory.

1 Introduction

Explicit termination forces an anonymous process to represent evidence that information has had enough time to propagate. This is different from stabilization: a binary consensus value can be propagated by a two-state monotone flooding process, while a process that must know when it is safe to stop needs additional local timing information.

The state-phase line of work studied this cost when every node receives the free common phase

$$\phi_t = t \bmod P$$

but not the absolute round number. In that model a direct predecessor proved, for known network size n ,

$$2 + \left\lceil \frac{\lfloor n/2 \rfloor}{P} \right\rceil \leq S_n(n, P) \leq 2 + \left\lceil \frac{n-1}{P} \right\rceil,$$

after improving an earlier split-cycle lower bound. The upper bound is a block-counter implementation of monotone zero flooding with horizon $n-1$. The remaining question is whether the factor-two gap in the lower numerator reflects the task or only the static-cycle witness.

This paper shows that the latter is largely responsible. Static cycles are convenient because their homogeneous executions are symmetric, but their ordinary diameter is only about $n/2$. The key observation is that static symmetry is unnecessary. If every snapshot is d -regular, then a homogeneous execution is blind to the edge identities: every node has the same state, broadcasts the same bit, and receives the same degree- d count vector. The adversary can therefore change the regular topology aggressively without changing the homogeneous state orbit.

We exploit this freedom by constructing two disjoint backward causal cones. Each cone has a constant-size first expansion forced by regularity and then grows by only one vertex per round. Explicit path/cycle-plus-matching completions fill the rest of every snapshot while preserving regularity and connectivity. The resulting waiting-sum lower bound is $n - O(1)$, which combines with two-orbit phase packing to give a near-exact state lower bound with leading constant one.

A second result isolates the smallest state budget. Three persistent states necessarily have the normal form “final zero, one live state, final one.” On a static path this collapses the pre-termination symmetry enough to determine exactly how much free phase is needed: three states are possible precisely when the period is at least $n-1$.

Main contributions.

1. A regular-homogeneous invariance lemma and a dynamic causal-cone construction that decouple homogeneous state evolution from mixed-input information propagation.

2. For all $n \geq 15$,

$$2 + \left\lceil \frac{n-13}{P} \right\rceil \leq S_n(n, P) \leq 2 + \left\lceil \frac{n-1}{P} \right\rceil.$$

3. For even $n \geq 12$, the stronger cubic lower bound

$$S_n(n, P) \geq 2 + \left\lceil \frac{n-10}{P} \right\rceil.$$

4. The exact three-state saturation threshold

$$S_n(n, P) = 3 \iff P \geq n-1 \quad (n \geq 4).$$

5. An explicit finite construction audit over 6253 complete parameter cases; the computation is used only to test the constructive lemmas, not as a substitute for proof.

2 Relation to prior work

The direct predecessor is Yonezu’s two-orbit packing note [2], which made the known-diameter state complexity exact and strengthened the known-size lower bound to $2 + \lceil \lfloor n/2 \rfloor / P \rceil$. The present paper addresses the residual known-size gap by replacing the static-cycle witness with regular dynamic witnesses and by isolating the exact three-state threshold.

Parzych and Daymude study memory lower bounds for terminating broadcast in anonymous, synchronous, 1-interval-connected dynamic networks [4]. Their results show that termination memory is a genuine resource in nearby anonymous dynamic settings, but the task, start semantics, and objective differ from the periodic-phase binary-consensus state complexity considered here.

Blanc, Di Luna, and Viglietta study deterministic computation when every anonymous node broadcasts one bit and observes only the numbers of neighboring zero and one broadcasts [3]. Their results establish broad computational power and round-complexity bounds under the same severe communication interface. They do not give the persistent state–phase thresholds studied here.

Di Luna and Viglietta develop finite-state and self-stabilizing computation in anonymous dynamic networks [5]. Their finite-state stabilization framework addresses general computation rather than exact persistent-state requirements for explicitly terminating binary consensus with an externally supplied phase. Turau gives a randomized $O(\log \log n)$ -memory algorithm for anonymous dynamic broadcast with stabilizing termination [6]; randomization, the broadcast task, and stabilizing rather than explicit termination place that result outside the present state-threshold question.

Modulo-clock synchronization is a different line. Penet de Monterno, Charron-Bost, and Merz synchronize clocks modulo P in dynamic networks [7], and Charron-Bost and Penet de Monterno study self-stabilizing clock synchronization [8]. Those works compute or recover a clock signal; here the common periodic phase is supplied externally for free, and the question is how many persistent states explicit consensus still requires.

A targeted literature search through September 6, 2026 did not identify a prior theorem giving either the near-exact $n - O(1)$ numerator bounds or the exact three-state threshold for the stated model. This is a scoped literature-screening statement, not a certification of novelty.

3 Model and definitions

We use the same state–phase model as the direct predecessor on periodic-time explicit consensus. Fix n and P . Nodes are anonymous, deterministic, and execute identical code. In propagation round t , the adversary chooses a simple undirected connected graph G_{t+1} . Each node broadcasts one bit to all current neighbors and observes only the pair (c_0, c_1) of received zero and one broadcasts. The common free phase is

$$\phi_t = t \bmod P.$$

The broadcast and transition functions have the form

$$\sigma : Q \times \{0, \dots, P-1\} \rightarrow \{0, 1\},$$

$$\delta : Q \times \{0, \dots, P-1\} \times \mathbb{N} \times \mathbb{N} \rightarrow Q.$$

The initialization is deterministic; write $\iota(b) \in Q$ for the persistent state of an input- b node at time 0. Finality and output depend only on persistent state. Once a final state is entered, the persistent state and output never change, although the node remains communication-present.

4 Dynamic causal locality and regular homogeneous orbits

For a target v at time T , define its backward causal sets by

$$B_T(v) = \{v\}, \quad B_t(v) = B_{t+1}(v) \cup N_{G_{t+1}}(B_{t+1}(v)) \quad (0 \leq t < T).$$

Lemma 1 (Dynamic causal-cone locality). *Fix a graph sequence and a target v at time T . If two executions have the same initial persistent states on $B_0(v)$, then v has the same state at every time $0, \dots, T$ in the two executions.*

Proof. Induct backward through the dependency relation. The state of a node at time $t+1$ depends only on its own state at time t , the common phase ϕ_t , and the broadcast bits of its neighbors in G_{t+1} . Thus every dependency of the target at time T is contained in the recursively defined backward cone. The phase sequence is common to the two executions, so equality of the initial states on $B_0(v)$ implies equality of all states and broadcasts that can influence v through time T . $\square$

Lemma 2 (Regular homogeneous invariance). *Fix $d \geq 1$. On every dynamic graph sequence whose snapshots are all d -regular, a homogeneous input execution has one common persistent state q_t at every node at each time t . Moreover, the augmented state (q_t, ϕ_t) evolves by a deterministic map*

$$G_d : Q \times \{0, \dots, P-1\} \rightarrow Q \times \{0, \dots, P-1\}$$

that is independent of the particular d -regular topology schedule.

Proof. If all nodes have the same state q at phase ϕ , they broadcast the same bit $\sigma(q, \phi)$. Every node has degree d , so every node receives the same count vector, either $(d, 0)$ or $(0, d)$. Determinism therefore gives the same next state at every node. The observation depends only on (q, ϕ, d) , not on the identities of the incident edges or on the topology schedule. $\square$

For $b \in \{0, 1\}$, let τ_b be the first final time in the homogeneous input- b execution on a d -regular schedule. By the preceding lemma, τ_b depends on the algorithm, n, P , and d , but not on the particular d -regular schedule.

Lemma 3 (Regular two-orbit packing). *For the two opposite homogeneous executions on d -regular schedules,*

$$P|Q| \geq \tau_0 + \tau_1 + 2P.$$

Proof. The proof is the same phase-augmented packing argument used for homogeneous static cycles. Before the first final time, an augmented state cannot repeat within either homogeneous orbit, or deterministic iteration of G_d would loop and prevent first termination. The two pre-final augmented orbits are disjoint: if they met, the two deterministic suffixes would coincide and eventually reach the same final output, contradicting opposite unanimity requirements. Finally, the distinct immutable final states reached by the two homogeneous executions each sweep through all P phase values. Hence the augmented state space contains at least $\tau_0 + \tau_1 + 2P$ distinct pairs. $\square$

5 Explicit regular protected blocks

Lemma 4 (Quartic protected block). *Let S be a vertex set of size $s \geq 5$ and let $x \notin S$. There is a simple connected partial graph on $S \cup \{x\}$ such that every vertex of S has degree 4,*

$$N(S) \setminus S = \{x\},$$

and x has exactly two incident edges into S . For $s = 5$, any prescribed target $z \in S$ can be chosen to have all four neighbors inside S .

Proof. Start from the square C_s^2 , which is 4-regular for $s \geq 5$. Delete one edge uv and add ux and vx . All vertices of S again have degree 4, while x has two incident edges into S . For $s = 5$, $C_5^2 = K_5$; choose the deleted edge disjoint from the prescribed target z . $\square$

Lemma 5 (Cubic protected block). *Let S be a vertex set of size $s \geq 4$ and let $x \notin S$. There is a simple connected partial graph in which every vertex of S has degree 3 and $N(S) \setminus S = \{x\}$. If s is even, x has two incident edges into S ; if s is odd, x has one incident edge into S . For $s = 4$, any prescribed target $z \in S$ can be chosen to have all three neighbors inside S .*

Proof. If s is even, take a cycle C_s and add the opposite-vertex perfect matching, pairing vertex i with $i + s/2$ (indices modulo s). These are non-cycle edges for $s \geq 4$, so the result is a connected cubic graph. Delete one matching edge uv and add ux, vx . For $s = 4$ this is K_4 with one edge replaced by the two edges to x , and the deleted edge can be chosen disjoint from the prescribed target.

If $s = 2h + 1$ is odd, take C_s and a matching

$$\{i(i+h) : 1 \leq i \leq h\}$$

on the vertices $1, \dots, 2h$, leaving vertex 0 unmatched. The matching edges are non-cycle edges because $h \geq 2$. Add the edge $0x$. Every vertex of S then has degree 3, and x has one edge into S . $\square$

6 Explicit completion of the complement

Lemma 6 (Quartic complement completion). *Let R be a vertex set with $|R| \geq 6$. Mark at most two vertices of R as connectors. Require each connector to have degree 2 inside R and every other vertex to have degree 4. Then there is a simple connected graph on R realizing those degrees.*

Proof. With no connector, use $C_{|R|}^2$.

With one connector x , write the remaining vertices as $o_1, \dots, o_q$, where $q \geq 5$. Use the cycle

$$x, o_1, o_2, \dots, o_q, x,$$

so every vertex currently has degree 2. Add a Hamiltonian cycle on the ordinary vertices that avoids the path edges $o_i o_{i+1}$. One explicit order is

$$o_1, o_3, o_5, \dots, o_2, o_4, o_6, \dots, o_1,$$

with the odd and even subsequences truncated in the natural way. For $q \geq 5$, none of its edges is a path edge. The ordinary vertices reach degree 4 while x remains degree 2.

With two connectors x, y and $|R| \geq 7$, use the base cycle

$$x, o_1, \dots, o_q, y, x$$

and the same edge-disjoint Hamiltonian cycle on the $q \geq 5$ ordinary vertices. Both connectors have degree 2 and all ordinary vertices degree 4.

The only remaining case is $|R| = 6$ with two connectors. Start from K_6 and delete the five edges

$$xy, xa, xb, yc, yd,$$

where a, b, c, d are the four ordinary vertices. The connectors then have degree 2 and all ordinary vertices degree 4. The resulting graph is connected. $\square$

Lemma 7 (Cubic complement completion). *Let R be a vertex set with $|R| \geq 5$. Mark at most two vertices as connectors, each with required degree 1 or 2 inside R , and require every ordinary vertex to have degree 3. If the total required degree is even, then a simple connected realization exists.*

Proof. The parity condition is necessary by the handshake lemma. We give explicit realizations.

With no connector, the parity condition makes $|R|$ even. Use a cycle plus a perfect matching on opposite vertices.

With one connector x of required degree 2, use the cycle

$$x, o_1, \dots, o_q, x$$

and add a perfect matching on the even number q of ordinary vertices, using non-path edges. If x has required degree 1, use a cycle on the odd number q of ordinary vertices, attach x to o_1 , and add a perfect matching on $o_2, \dots, o_q$ using non-cycle edges.

With two connectors of required degrees $(1, 1)$, use the path

$$x, o_1, \dots, o_q, y$$

and add a perfect matching on the even number q of ordinary vertices. For degrees $(2, 2)$, use the cycle

$$x, o_1, \dots, o_q, y, x$$

and again add a perfect matching on the even number q of ordinary vertices. For degrees $(1, 2)$ and $q \geq 5$, use the cycle

$$y, o_1, o_2, \dots, o_q, y,$$

attach x to o_1 , and add a perfect matching on $o_2, \dots, o_q$. The parity condition makes q odd.

In every matching step above, the vertices to be matched form an even ordered list $w_1, \dots, w_m$ with $m \geq 4$, except for the special case below. Pair w_i with $w_{i+m/2}$ for $1 \leq i \leq m/2$. Each pair is separated by at least two positions in the displayed path or cycle, so no matching edge duplicates an existing edge. The only size-five case not covered by these formulas is $q = 3$ with connector degrees $(1, 2)$. On vertices x, y, a, b, c , take K_5 and delete

$$xy, xa, xb, yc.$$

The remaining degrees are respectively 1, 2, 3, 3, 3, and the graph is connected. $\square$

7 Slow disjoint causal cones

Proposition 1 (Quartic slow-cone schedule). *Let $n \geq 15$ and let $t_0, t_1 \geq 0$ satisfy*

$$t_0 + t_1 \leq n - 14.$$

There exists a sequence of connected 4-regular graphs and two targets z_0, z_1 such that their backward causal cones through horizons t_0, t_1 are disjoint at time 0.

Proof. Choose disjoint vertex pools U_b of size $t_b + 4$, choose a target $z_b \in U_b$ for each $b \in \{0, 1\}$, and let

$$M = V \setminus (U_0 \cup U_1).$$

The hypothesis gives $|M| \geq 6$.

For each b with $t_b > 0$, construct nested sets

$$C_b(t_b) = \{z_b\},$$

$$|C_b(t_b - 1)| = 5,$$

and, for $0 \leq t \leq t_b - 2$,

$$C_b(t) = C_b(t + 1) \cup \{x_{b,t}\},$$

using fresh vertices from U_b . Thus $|C_b(0)| = t_b + 4$.

We now choose the snapshot G_{t+1} independently for each round t . For each active target b with $t < t_b$, impose one protected block.

If $t = t_b - 1$, apply the quartic protected-block lemma to $S = C_b(t)$. Choose its external connector in M , distinct from the connector of any other protected block in the same snapshot, and choose the deleted edge disjoint from z_b . Then all four neighbors of z_b lie in $C_b(t)$, so the one-step backward predecessor of $\{z_b\}$ is exactly $C_b(t)$.

If $t \leq t_b - 2$, apply the protected-block lemma to $S = C_b(t + 1)$ with external connector $x_{b,t} = C_b(t) \setminus C_b(t + 1)$. Then

$$C_b(t + 1) \cup N_{G_{t+1}}(C_b(t + 1)) = C_b(t).$$

The protected sets are disjoint, and their connectors are distinct and lie outside both protected sets. Remove the protected-set vertices from consideration and let R be the remaining vertices, including the at most two connectors. Because each protected set lies inside its pool,

$$|R| \geq |M| \geq 6.$$

Each connector already has two edges into its protected block and therefore needs degree 2 inside R ; every other vertex of R needs degree 4. Complete R by the quartic complement-completion lemma. The union is simple, connected, and 4-regular.

For every positive horizon, induction on the backward-cone recursion gives that the exact cone of z_b at time t is $C_b(t)$ for every relevant t . When $t_b = 0$ the time-zero cone is simply $\{z_b\}$. Hence in all cases the two time-zero cones are contained in the disjoint pools U_0, U_1 . $\square$

Proposition 2 (Cubic slow-cone schedule). *Let $n \geq 12$ be even and let $t_0, t_1 \geq 0$ satisfy*

$$t_0 + t_1 \leq n - 11.$$

There exists a sequence of connected 3-regular graphs and two targets whose time-zero backward causal cones through horizons t_0, t_1 are disjoint.

Proof. Choose disjoint pools U_b of size $t_b + 3$, choose a target $z_b \in U_b$ for each $b \in \{0, 1\}$, and let $M = V \setminus (U_0 \cup U_1)$, so $|M| \geq 5$. For each b with $t_b > 0$, construct nested sets exactly as above, except that the first backward expansion has size 4 and each earlier round adds one vertex. At a first-expansion round, choose the external connector in M , distinct from any other connector in that snapshot; at earlier rounds use the fresh pool vertex that is added to the nested cone.

Use the cubic protected-block lemma for each active target. A connector has two incident block edges when the protected-set size is even and one incident block edge when that size is odd, so it requires residual degree 1 or 2 inside the complement. All non-connectors require degree 3 there. At every round the complement has at least five vertices. It remains only to check the parity condition in the cubic complement-completion lemma. Let S be the union of the protected blocks and let e_{cut}

be the number of already fixed block-to-connector edges. Since every vertex of S already has degree 3,

$$3|S| = 2|E(S)| + e_{\text{cut}},$$

so

$$e_{\text{cut}} \equiv |S| \pmod{2}.$$

The required degree sum inside the complement $R = V \setminus S$ is

$$3|R| - e_{\text{cut}} \equiv |R| - |S| \equiv n \equiv 0 \pmod{2},$$

because n is even. Hence the cubic complement-completion lemma applies.

As in the quartic case, induction on the backward-cone recursion gives the prescribed nested sets exactly for positive horizons, while a zero-horizon cone is the singleton target. The two time-zero cones are therefore disjoint. $\square$

8 Near-exact known-size lower bounds

Theorem 1 (Quartic known-size lower bound). *For every $n \geq 15$ and $P \geq 1$,*

$$S_n(n, P) \geq 2 + \left\lceil \frac{n-13}{P} \right\rceil.$$

Proof. Fix a correct algorithm with state set Q , and let τ_0, τ_1 be the two homogeneous first-final times on 4-regular schedules.

If $\tau_0 + \tau_1 \leq n - 14$, the quartic slow-cone proposition gives a connected 4-regular schedule with two targets whose time-zero backward cones are disjoint. Assign input 0 to every node in the first cone and input 1 to every node in the second cone; assign the remaining inputs arbitrarily. By dynamic causal-cone locality, target b reproduces its homogeneous input- b history through time τ_b and therefore reaches an immutable final output b . The two opposite final decisions coexist, contradicting agreement. Hence

$$\tau_0 + \tau_1 \geq n - 13.$$

Regular two-orbit packing now gives

$$P|Q| \geq n - 13 + 2P,$$

which is equivalent to the stated bound. $\square$

Theorem 2 (Even-size cubic refinement). *For every even $n \geq 12$ and $P \geq 1$,*

$$S_n(n, P) \geq 2 + \left\lceil \frac{n-10}{P} \right\rceil.$$

Proof. Use the cubic slow-cone proposition. Correctness implies

$$\tau_0 + \tau_1 \geq n - 10$$

for the two homogeneous first-final times on cubic schedules. Applying regular two-orbit packing yields

$$P|Q| \geq n - 10 + 2P,$$

and hence the result. $\square$

Combining the quartic lower bound with the existing block-counter upper bound gives, for $n \geq 15$,

$$2 + \left\lceil \frac{n-13}{P} \right\rceil \leq S_n(n, P) \leq 2 + \left\lceil \frac{n-1}{P} \right\rceil.$$

For even $n \geq 12$, the left side improves by replacing $n-13$ with $n-10$.

9 Exact three-state phase threshold

Lemma 8 (Three-state normal form). *Suppose a correct algorithm has exactly three persistent states. Then, after possibly exchanging the names 0 and 1, its states may be written*

$$Q = \{F_0, L, F_1\},$$

where F_0, F_1 are distinct immutable final states with outputs 0, 1, L is the unique nonfinal state, and

$$\iota(0) = F_0, \quad \iota(1) = L.$$

Proof. Unanimity requires that some reachable final state output 0 and some distinct reachable final state output 1. Thus at least two states are final. A correct mixed execution cannot start with both input values already in opposite immutable final states, so a nonfinal state is also necessary. With only three states, there are therefore exactly two relevant final states and exactly one nonfinal state L .

An input- b node cannot initialize in the opposite final state, because that would already violate unanimity in the homogeneous input- b execution. If both input values initialized in L , their homogeneous executions would be identical and could not terminate with opposite outputs. If both initialized in their respective final states, a mixed input would start with opposite immutable decisions. Hence exactly one input initializes final and the other initializes in L . Exchange the bit names if necessary. $\square$

Lemma 9 (Path symmetry before the first one-finalization). *Run a three-state algorithm in the all-one execution on the static path*

$$v_0 - v_1 - \cdots - v_{n-1}.$$

Before the first transition into F_1 , every node is in state L . At every such time, the two endpoints have identical local observations and all interior vertices have identical local observations. Consequently the first F_1 transition, when it occurs, is simultaneous on the entire endpoint class, the entire interior class, or both.

Proof. No node may enter F_0 in the all-one execution, because F_0 is immutable with output 0. Since L is the only nonfinal state, every node therefore remains in L until the first F_1 transition. The two endpoints each see one L neighbor, while every interior node sees two L neighbors. The phase is common and the transition rule is anonymous and deterministic. $\square$

Lemma 10 (Forced maximal zero wave at the boundary). *Assume the three-state normal form and suppose that, in the all-one execution on P_n , the first final-one event is interior-only at time*

$$P = n - 2.$$

In the mixed execution with v_0 initially zero and all other inputs one, correctness forces, for every $0 \leq j \leq P - 2$,

$$q_{v_j}(j) = F_0, \quad q_{v_{j+1}}(j) = L,$$

and the propagation-round- j transition at the frontier satisfies

$$\delta(L, \phi_j, \text{one broadcast from } F_0 \text{ and one broadcast from } L) = F_0.$$

Consequently

$$\sigma(F_0, \phi_j) \neq \sigma(L, \phi_j) \quad (0 \leq j \leq P - 2).$$

Moreover the same inequality also holds at phase $P - 1$.

Proof. The zero source v_0 is permanently in F_0 , and correctness forbids any node from entering F_1 in the same mixed execution. The only remaining state is therefore L .

In the all-one execution, v_{n-2} enters F_1 at time $P = n - 2$. For the mixed execution to avoid that same transition, the causal difference originating at v_0 must affect the round- $P - 1$ observation of v_{n-2} . The distance from v_0 to v_{n-3} is exactly

$$n - 3 = P - 1.$$

One-hop-per-round causality leaves no slack: a delay in any one of rounds $0, \dots, P - 2$ cannot be recovered later. Hence at time j the influence frontier must already have reached v_j , and because F_1 is forbidden, the differing state there must be F_0 . The next vertex v_{j+1} is still outside the causal influence and remains in L at the beginning of round j . To advance the frontier, its next state must be F_0 . This proves the displayed transition.

If the two broadcasts at phase ϕ_j were equal, the frontier vertex v_{j+1} would receive exactly the same count vector as its all-one counterpart and could not make the required different transition. Thus the broadcasts differ for $0 \leq j \leq P - 2$.

At the last phase $P - 1$, the vertex v_{n-2} begins the round in L . Its neighbors are $v_{n-3} \in F_0$ and $v_{n-1} \in L$. If the two broadcasts were equal at that phase, its received count vector would again equal the all-one count vector, forcing the forbidden transition into F_1 . Hence the inequality also holds at phase $P - 1$. $\square$

Theorem 3 (Exact three-state threshold). *For every $n \geq 4$ and $P \geq 1$,*

$$S_n(n, P) = 3 \iff P \geq n - 1.$$

Proof. Any correct algorithm needs at least three persistent states: unanimity requires distinct immutable final states for outputs 0 and 1, while mixed-input correctness requires a nonfinal state. The existing block-counter construction with horizon $H = n - 1$ uses one live state and two final states when $P \geq n - 1$. Hence $S_n(n, P) = 3$ in that regime.

For the converse, suppose that a correct three-state algorithm exists with $P \leq n - 2$. Put it in the normal form $\{F_0, L, F_1\}$.

Consider the all-one execution on the static path P_n , and let T be the first time at which some node enters F_1 . If no F_1 appears during the first P propagation rounds, then at time P every node is again in L and the phase is again 0. The entire global configuration has repeated, so the execution would never terminate. Hence

$$T \leq P \leq n - 2.$$

If an endpoint is among the first nodes entering F_1 , put a single zero at the opposite endpoint. Its distance is $n - 1 > T$, so the first endpoint has exactly the same local history through time T as

in the all-one execution and enters F_1 , while the zero endpoint is already in the immutable state F_0 . This violates agreement.

Thus the first event is interior-only. By path symmetry all interior vertices enter F_1 simultaneously. If $T < n - 2$, put a single zero at v_0 . The vertex v_{n-2} is at distance $n - 2 > T$ from that zero, so its local history through time T is unchanged and it enters F_1 , again conflicting with the immutable F_0 at v_0 . Therefore the only remaining case is

$$T = P = n - 2,$$

with all interior vertices entering F_1 at time P and both endpoints still in L .

For each phase ϕ , define the broadcast bits

$$a_\phi = \sigma(F_0, \phi), \quad \ell_\phi = \sigma(L, \phi), \quad b_\phi = \sigma(F_1, \phi).$$

Apply the forced maximal zero-wave lemma. In terms of the broadcast notation just introduced, it gives

$$a_\phi \neq \ell_\phi \quad \text{for every } \phi \in \{0, \dots, P - 1\},$$

and, for phases $0, \dots, P - 2$, it also gives the concrete frontier transition $L \rightarrow F_0$ when an interior live node sees one F_0 neighbor and one L neighbor.

Return to the all-one execution. At time P every interior vertex is in the immutable state F_1 , while the two endpoints are still in L . Termination requires each endpoint eventually to enter F_1 . Let $r \geq P$ be a propagation round at whose end an endpoint first enters F_1 , and put

$$\psi = r \bmod P.$$

At the beginning of round r that endpoint is in L and its unique neighbor is in F_1 . During the first phase- ψ round of the execution, the same endpoint was in L with a unique L neighbor and remained in L . Therefore

$$b_\psi \neq \ell_\psi.$$

Since the broadcast alphabet is binary and also $a_\psi \neq \ell_\psi$, necessarily

$$a_\psi = b_\psi.$$

Finally, take a static path and choose a target endpoint u . Place a single zero at the vertex whose distance from u is $\psi + 1$; all other inputs are one. The forced frontier transitions above propagate F_0 toward u at one edge per round during phases $0, \dots, \psi - 1$. Hence at the beginning of propagation round ψ , the unique neighbor of u is in F_0 , while u itself is still in L . The endpoint therefore sees the same received bit $a_\psi = b_\psi$ that caused an all-one endpoint with an F_1 neighbor to enter F_1 at phase ψ . Determinism forces u into F_1 . The source zero remains permanently in F_0 , contradicting agreement.

Hence no three-state algorithm exists when $P \leq n - 2$. □

10 Construction audit

The graph lemmas above are explicit and constitute the proof. As a separate implementation audit, the constructions were independently instantiated over finite parameter ranges and their claimed invariants were recomputed from the generated snapshots.

For quartic schedules, every ordered pair (t_0, t_1) with

$$15 \leq n \leq 40, \quad t_0, t_1 \geq 0, \quad t_0 + t_1 \leq n - 14$$

was checked, for a total of 3653 cases. For cubic schedules, every ordered pair with

$$12 \leq n \leq 40, \quad n \text{ even}, \quad t_0, t_1 \geq 0, \quad t_0 + t_1 \leq n - 11$$

was checked, for a total of 2600 cases. Across all 6253 complete schedules, every snapshot was simple, connected, and exactly regular of the intended degree; recomputed backward causal cones matched the prescribed nested cones; and the two time-zero cones were disjoint. No construction failure was observed.

These checks are not part of the mathematical proof and do not establish the theorem for unbounded n . Their purpose is to test the indexing and explicit graph constructions used in the proof.

11 Consequences and remaining gap

Whenever $n/P \rightarrow \infty$ (in particular, for fixed or sufficiently slowly growing P), the quartic theorem and the block-counter upper bound give

$$S_n(n, P) = \frac{n}{P} + O(1)$$

with leading constant one. At $P = 1$ the general bounds become

$$n - 11 \leq S_n(n, 1) \leq n + 1 \quad (n \geq 15),$$

while even $n \geq 12$ improves the lower side to

$$n - 8 \leq S_n(n, 1) \leq n + 1.$$

The exact three-state result identifies the first saturation boundary:

$$P_{\min}^{(3)}(n) = n - 1.$$

It does not by itself imply the full feasible-region formula for larger state budgets. In particular, we leave open whether

$$S_n(n, P) = 2 + \left\lceil \frac{n - 1}{P} \right\rceil$$

holds for all n and P , or whether dynamic topology can be used by algorithms as a distributed spatial memory to beat the block-counter upper construction for intermediate state budgets.

12 Claim boundary and limitations

The paper does not claim novelty for finite-state pumping, deterministic trajectory packing in general, causal-cone locality, regular graph constructions, monotone flooding, periodic clocks, or generic time-space tradeoffs. The claimed contribution is the task-specific use of regular dynamic witnesses to sharpen the known-size state lower bound to within an additive constant of the existing upper numerator, together with the exact three-state phase threshold.

The model assumes deterministic anonymous synchronous leaderless 1-interval-connected dynamic networks; a free globally aligned period- P phase; output and finality encoded in persistent state rather than in the phase; final nodes remaining present in communication; and one-bit broadcast-counting communication. Randomization, identifiers, ports, leaders, Byzantine faults, asynchronous execution, and disappearance-on-finalization semantics are outside scope.

13 Conclusion

Regularity lets a dynamic adversary preserve the homogeneous state orbit while changing the geometry of causal influence. This separates two roles that were entangled in the previous static-cycle lower bound: symmetry of the unanimous executions and propagation distance in a mixed execution. Explicit quartic and cubic schedules then force the sum of the two homogeneous first-final times to be $n - O(1)$, and two-orbit phase packing converts that waiting into a near-exact persistent-state lower bound.

At the minimum state budget, the structure is sharper. Three states leave only one live state, and on a static path this forces a maximal-speed zero wave at the critical boundary. The one-bit alphabet then makes final-zero and final-one communication indistinguishable at some phase, yielding the exact threshold $P = n - 1$.

The remaining problem is no longer a factor-two uncertainty. It is whether the final additive gap can be closed for arbitrary state budgets.

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